Predicting the human interactome

Predicting protein-protein interactions in the human proteome

Jing Zhang et al. — *Science* 390, eadt1630 (2025)

DOI: 10.1126/science.adt1630
Published online 25 September 2025; issue dated 23 October 2025.

Deeper evolutionary alignments + fast learned screening + structural hypotheses.

The question: which human proteins directly interact?

A discovery problem with an extreme class imbalance
Roughly 20,000 proteins create 200 million possible heteromeric pairs. The estimated interactome contains only 74,000–200,000 true interactions.

1. Experimental screens are incomplete and condition dependent; databases disagree.
2. Accurate complex structure prediction is too expensive for every pair.
3. Human coevolutionary signals are weak when few animal orthologs are available.

Goal: identify interactions at high precision, then provide structural hypotheses.

Introduction; Fig. 1A. Heteromeric means two different protein identities.

Two bottlenecks, two complementary interventions

More evolutionary information
Assemble protein sequences directly from draft genomes and raw reads. Build deeper **omicMSAs**, with orthologs from 21,414 eukaryotic species.

More training examples, cheaper inference
Distill domain–domain contacts from predicted monomers. Train **RF2-PPI**, a fast pair classifier using MSA and residue-pair features.

Fast screens narrow the search; AlphaFold2 scores and models the survivors.

Main output: 17,849 PPIs at an *estimated* 90% precision; 3,631 are absent from the reference experimental PPI databases.

Abstract; Figs. 1–3. Novelty here is database-relative, not proof of no previous report.

Running example: pair orthologs before looking for coevolution

For human proteins A and B, concatenate orthologs from the *same species*:

Species	A: candidate interface site	B: candidate interface site
Human	Lys (+)	Glu (−)
Mouse	Lys (+)	Asp (−)
Fish	Arg (+)	Glu (−)
Insect	Glu (−)	Lys (+)

Intuition: compensating substitutions may preserve a contact. More correctly paired taxa give more opportunities to distinguish contact constraints from noise.

This table is a teaching example, not a reported sequence alignment. Ancestry and incorrect orthologs can also mislead.

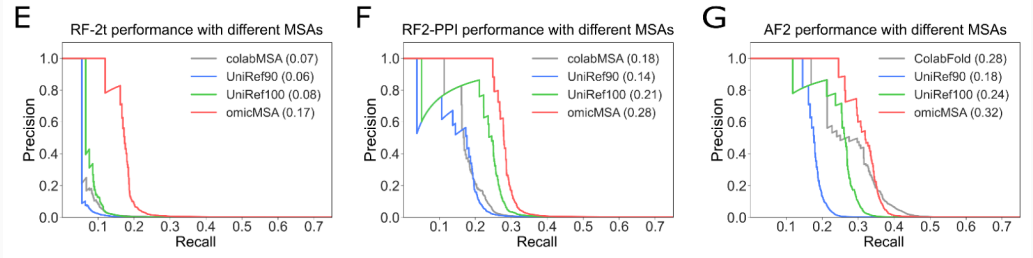
omicMSA: recover coding information beyond annotated proteins

1. Mine NCBI Genome and Sequence Read Archive; choose one representative genomic dataset per species.
2. Assemble coding sequences with reference proteins and splice-aware alignment, including draft genomes and genome/transcriptome reads.
3. Align predicted sequences to human proteins; use reciprocal best hits to distinguish orthologs from paralogs.
4. Pair orthologs by species to build the complex input.

Why this matters specifically for humans
Only 20% of available eukaryotic genomes had annotated proteins (June 2024). Animal sequence recovery complements metagenomic databases.

Main text, "Harnessing untapped genomic sequence data"; Fig. 1B–D. Detailed M2 supplement inaccessible.

Evidence for better inputs: the same predictors improve



Comparison: the same three predictors receive different MSAs. Red curves use omicMSAs.

Result: omicMSAs improve discrimination for all three predictors.

Interpretation: better input information contributes independently of architecture.

Fig. 1E–G, reproduced crop. Precision is reweighted to a 1:1000 signal-to-noise ratio.

Why conventional benchmark precision is misleading

The benchmark has 3,000 positive pairs and 30,000 random negatives: a 1:10 ratio. A proteome-wide screen is closer to 1:1000.

$$\widehat{\text{precision}} = \frac{0.01\,\text{TP}}{0.01\,\text{TP} + \text{FP}}, \qquad \text{recall} = \frac{\text{TP}}{3000}.$$

Teaching calculation: a threshold recovering 1,000 positives and 10 negatives looks 99% precise in the sampled data. After weighting, precision is $10/(10 + 10) = 50\%$.

A tiny false-positive rate can still swamp the true signal. High-precision performance is the relevant criterion for de novo discovery.

Fig. 1A,E–G; arithmetic example constructed for teaching. TP/FP refer to benchmark labels.

Training signal: domains within a monomer can mimic a PPI

Self-distillation assumption
Interfaces between domains of a single protein share evolutionary and physical constraints with interfaces between separate proteins.

1. Start from 53.7 million AFDB models filtered at 50% sequence identity.
2. Identify domains using structure, predicted aligned error (PAE), and InterPro annotation.
3. Keep confident contacts: at least 25 residue contacts within 6 Å; mean inter-domain PAE < 8.
4. Cluster at 30% identity and remove close human homologs: **223,441 DDI clusters**, versus 13,231 PDB PPI clusters.

Benefit: a 17-fold larger positive training resource. **Risk:** covalently linked domains do not represent every transient interaction.

Main text; Fig. 2A–B. DDI: domain–domain interaction.

RF2-PPI learns both contact geometry and non-interaction

Inputs and computation

1. Protein MSAs, including paired MSAs.
2. Twelve Evoformer-like blocks update MSA and pair representations.
3. Predict residue distances/orientations and interaction evidence.

Training mixture

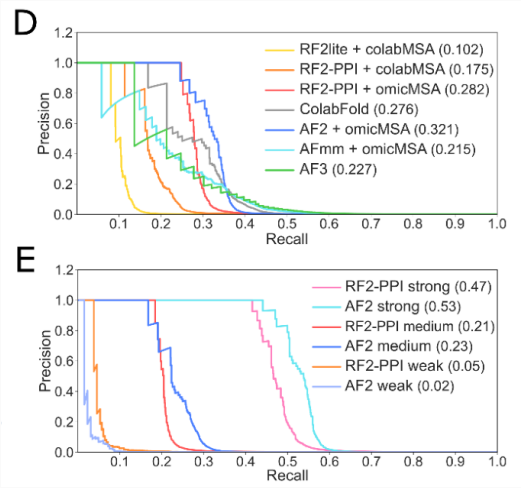
Type	Relative weight
Monomers	2
Positive DDIs	1
Negative DDIs	1
Positive PPIs	1
Negative PPIs	1

Unlike full RF2, the final model omits the integrated 3D structure track. Losses supervise masked tokens, pair distances and orientations.

Authors' rationale: negatives teach rejection; reducing 3D emphasis may help detect weak, motif-mediated interactions.

Fig. 2C and main text. Ablations reported in S17–S18 were not directly accessible.

Better discrimination is different from better complex folding



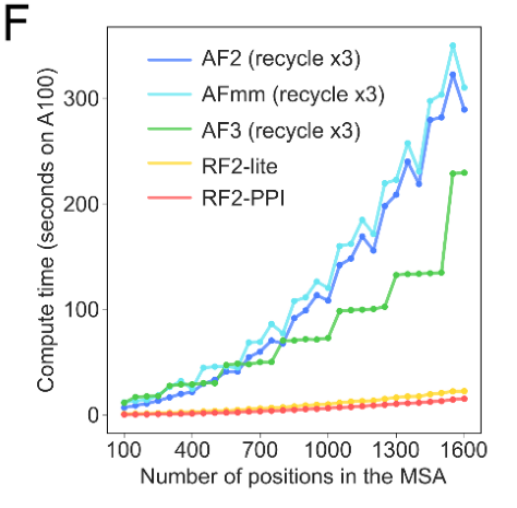
RF2-PPI improves AUCPR 1.7-fold over RF2-lite. With omicMSAs, it slightly exceeds the tested ColabFold pipeline.

AF2 + omicMSA performs best for the de novo discrimination task. RF2-PPI is the economical first pass.

AF-multimer and AF3 fare worse at **1:1000 prevalence**, despite their strength at modeling true complexes.

Fig. 2D–E crop. E stratifies by interface size. Model rankings depend on inputs, scoring and prevalence.

Speed makes the cascade feasible; weak interfaces remain difficult



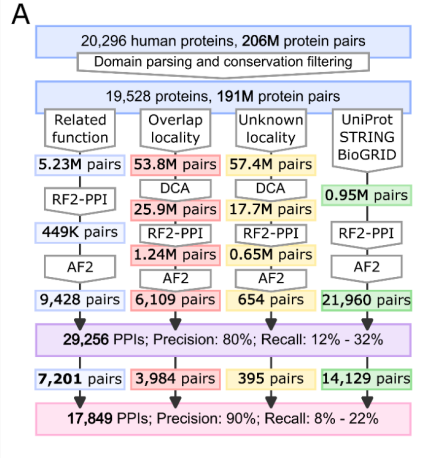
Compute: RF2-PPI is approximately 20 times faster than AF2, and 50% faster than RF2-lite.

Interface dependence: both methods perform better for larger interfaces. RF2-PPI has an advantage over AF2 for the weak-interface benchmark.

Interface size is a proxy for stable versus transient binding, not a direct affinity measurement.

Fig. 2F crop; Fig. 2E and main text. Runtime depends on protein length and configuration.

The screen: spend expensive inference only on survivors



19,528 retained proteins yield 191 million pairs. Colocalized pairs and pairs with unknown localization enter the de novo branch.

1. DCA prefilter: 111.2M $\rightarrow$ 43.6M pairs.
2. RF2-PPI: retain 1.9M pairs with interaction score > 0.3.
3. AF2: final scoring and complex modeling.

De novo branch: **4,379** PPIs at estimated 90% precision (6,763 at 80%).

Fig. 3A crop; main text. DCA = direct coupling analysis. Some branches overlap; counts are not additive.

Prior evidence changes the threshold, not the definition of a PPI

Candidate source	Input pairs	90% set
De novo, no interaction prior	111.2M	4,379
STRING genetic association	5.23M	7,201
Prior physical-interaction evidence	1.15M	14,129
Union after removing overlaps		**17,849**

A genetic association

can raise the prior probability of binding, but it is *not* evidence of direct physical contact by itself.

The same nominal score has different precision in different candidate pools. Random pairs require thresholds near 0.99; evidence-enriched pools can use lower cutoffs.

Main text; Fig. 3A,C–D. "90%" is estimated from calibrated benchmarks, not prospective validation.

The resulting resource is substantial, but far from complete

Estimated precision	Predicted PPIs	Estimated recall
80%	29,257	12–32%
90%	17,849	8–22%

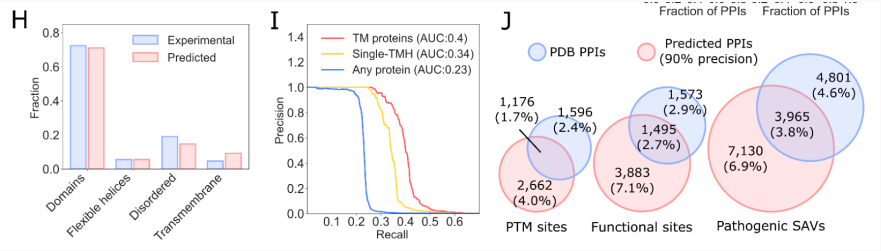
Within the 90% set:

1. 3,037 have orthologous PDB templates.
2. 11,181 have other experimental-database support.
3. 3,631 are absent from those experimental PPI databases.

Recall ranges inherit uncertainty in the total human interactome size. A pair missing from the resource cannot be treated as non-interacting.

Main text; Fig. 4A. High-confidence set is the focus of downstream biology.

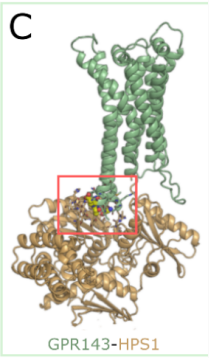
Which parts of the interactome does the pipeline favor?



1. 71% of predicted interfaces are primarily globular-domain mediated.
2. Disordered-linker interactions are underrepresented; poorly conserved flexible sequence was removed to reduce false positives.
3. Membrane-protein benchmarks support utility for transmembrane interactions, a difficult class for many experiments.

Fig. 3H–J crop: interface classes, membrane-protein PR, and interface annotation counts.

Disease interpretation: a structural hypothesis is a starting point



Example: GPR143–HPS1. Both proteins function in melanosome biology and are associated with albinism.

Four GPR143 variants associated with ocular albinism type 1 lie at the predicted interface (yellow spheres).

Authors' proposal: loss of this interaction could contribute to disease.

What remains to test: binding in the correct cellular compartment, loss of binding after mutation, and rescue of the relevant phenotype.

Fig. 4C crop; main text. Variant colocalization supports prioritization, not causal proof.

Unknown proteins: use partners to propose a focused experiment

Example: C8orf82 and electron transfer flavoprotein
A predicted association with the ETFA–ETFB complex suggests a role in assembly, stability, or metabolic regulation.

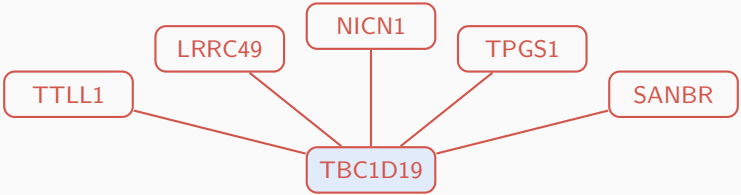
The study reports 1,413 predicted PPIs involving proteins of unknown function. More than half of the 2,700 least-characterized PPIs involve transmembrane proteins.

Our proposed follow-up: perturb C8orf82; measure ETF assembly, protein abundance and enzymatic activity; compare an interface mutant with wild type rescue.

Discriminating outcome: selective loss of ETF assembly would support a scaffold/chaperone role; preserved assembly with altered activity would favor regulation.

Main text; Fig. 4H. Follow-up design is journal-club interpretation, not a reported experiment.

Pairwise edges can suggest larger complexes



404 putative complexes; 139 candidate additions to known complexes. For tubulin polyglutamylase, proposed additions are TBC1D19, CSTPP1 and SANBR.

Compatible pairwise models suggest assembly hypotheses; they do not establish simultaneous occupancy or stoichiometry.

Adapted from Fig. 5A and main text. Schematic shows five reported TBC1D19 contacts, not the complete complex.

How strong is the claimed 90% precision?

1. **Positive controls are selected for agreement.** The intersection of three databases may favor well-studied, stable and easy-to-detect interactions.
2. **Random negatives are not experimentally proven negatives.** Unknown true interactions can contaminate the negative set; easy negatives may not represent all realistic alternatives.
3. **Precision depends on the assumed base rate.** Reweighting corrects prevalence only if benchmark score distributions transfer to the screen.

The reported database-precision estimates also use computational recovery and model recall. They are *model-dependent estimates*, not independent experimental error audits.

Benchmark construction and "Integration ... knowledge bases"; Fig. 3F. Critique is our interpretation.

What the study does not establish

1. **Completeness:** 4.8% of proteins are excluded, many linker residues are omitted, and early filters can lose positives. Heteromeric pairs are the focus.
2. **Cellular context:** binding can depend on expression, isoform, modification, tissue, localization, ligand or assembly state.
3. **Prospective biological validation:** structural examples and variant mapping motivate experiments; this study's central evidence is computational benchmarking.

Access qualification for these notes
The complete main article and figure captions were read. The supplementary PDF could not be retrieved; detailed M2/M4/M5 calibration and ablation procedures were not independently audited here.

Main text and Figs. 1–5; source/access log in README.md.

Discussion: what would make this resource convincing for your project?

1. Choose candidates by **score + input coverage + cellular plausibility**, and inspect whether the contact lies in a retained domain or flexible region.
2. Validate a balanced panel: novel membrane interactions, weak-interface pairs, and similarly scored controls; use an orthogonal binding assay.
3. Test mechanism through interface mutations and functional rescue, while measuring expression and localization.

Take-home: recovering unused evolutionary information and distilling predicted domain contacts make a human-scale PPI screen practical. The product is a high-value hypothesis resource with substantial blind spots.

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